

EX NO - 1	SETTING UP THE ENVIRONMENT AND PREPROCESSING THE DATA
23/06/2026	

To set up a fully functional machine learning development environment and to perform data preprocessing operations like handling missing values, encoding categorical variables, feature scaling, and splitting datasets.

1. Install Required Libraries:
 - a. Install numpy, pandas, matplotlib, seaborn, and scikit-learn using pip.
2. Import Libraries.
3. Load Dataset:
 - a. Load any dataset (e.g., Titanic or Iris) using pandas.
4. Data Exploration:
 - a. Use `df.info()`, `df.describe()`, `df.isnull().sum()` to understand the data.
5. Handle Missing Values:
 - a. Use `.fillna()` or `.dropna()` depending on the strategy.
6. Encode Categorical Data:
 - a. Use `pd.get_dummies()` or `LabelEncoder`.
7. Feature Scaling:
 - a. Normalize or standardize the numerical features using `StandardScaler` or
8. `MinMaxScaler`.
9. Split Dataset:
 - a. Use `train_test_split()` from `sklearn` to create training and testing sets.
10. Display the Preprocessed Data.

CODE:

```
1 # =====
2 # EXPERIMENT 1
3 # SETTING UP THE ENVIRONMENT AND PREPROCESSING THE DATA
4 # =====
5
6 # Import libraries
7 import pandas as pd
8 import numpy as np
9 import seaborn as sns
10 import matplotlib.pyplot as plt
11
12 from sklearn.model_selection import train_test_split
13 from sklearn.preprocessing import LabelEncoder, StandardScaler
14
15 # -----
16 # Load Dataset
17 # -----
18 df = sns.load_dataset("titanic")
19
20 print("First 5 Rows:")
21 print(df.head())
22
23 # -----
24 # Dataset Information
25 # -----
26 print("\nDataset Information")
27 print(df.info())
28
29 print("\nStatistical Summary")
30 print(df.describe())
31
32 print("\nMissing Values")
33 print(df.isnull().sum())
34
35 # -----
36 # Handle Missing Values
37 # -----
38
39 # Fill age with median
40 df["age"] = df["age"].fillna(df["age"].median())
41
42 # Fill embark_town with mode
43 df["embark_town"] = df["embark_town"].fillna(
44     df["embark_town"].mode()[0]
45 )
46
47 # Remove deck column (too many missing values)
48 df = df.drop(columns=["deck"])
49
50 # -----
51 # Encode Categorical Variables
52 # -----
53
54 le_sex = LabelEncoder()
55 df["sex"] = le_sex.fit_transform(df["sex"])
56
57 le_town = LabelEncoder()
58 df["embark_town"] = le_town.fit_transform(df["embark_town"])
59
60 # -----
61 # Drop Unnecessary Columns
62 # -----
63
64 df = df.drop(
65     columns=[
66         "embarked",
67         "class",
68         "who",
69         "alive",
70         "adult_male",
71         "alone"
72     ]
73 )
74
75 # -----
76 # Feature Scaling
77 # -----
78
79 scaler = StandardScaler()
80
81 numerical_columns = ["age", "fare"]
82
83 df[numerical_columns] = scaler.fit_transform(df[numerical_columns])
84
85 # -----
86 # Split Dataset
87 # -----
88
89 X = df.drop("survived", axis=1)
90 y = df["survived"]
91
92 X_train, X_test, y_train, y_test = train_test_split(
93     X,
94     y,
95     test_size=0.20,
96     random_state=42
97 )
98
99 # -----
100 # Display Results
101 # -----
102
103 print("\nTraining Data Shape:", X_train.shape)
104 print("Testing Data Shape :", X_test.shape)
105
106 print("\nFirst 5 Rows of Training Data")
107 print(X_train.head())
```

OUTPUT:

First 5 Rows:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

Dataset Information

<class 'pandas.DataFrame'>

RangeIndex: 891 entries, 0 to 890

Data columns (total 15 columns):

#	Column	Non-Null Count	Dtype
0	survived	891 non-null	int64
1	pclass	891 non-null	int64
2	sex	891 non-null	str
3	age	714 non-null	float64
4	sibsp	891 non-null	int64
5	parch	891 non-null	int64
6	fare	891 non-null	float64
7	embarked	889 non-null	str
8	class	891 non-null	category
9	who	891 non-null	str
10	adult_male	891 non-null	bool
11	deck	203 non-null	category
12	embark_town	889 non-null	str
13	alive	891 non-null	str
14	alone	891 non-null	bool

dtypes: bool(2), category(2), float64(2), int64(4), str(5)

memory usage: 80.7 KB

None

Statistical Summary

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

First 5 Rows of Training Data

	pclass	sex	age	sibsp	parch	fare	embark_town
331	1	1	1.240235	0	0	-0.074583	2
733	2	1	-0.488887	0	0	-0.386671	2
382	3	1	0.202762	0	0	-0.488854	2
704	3	1	-0.258337	1	0	-0.490280	2
813	3	0	-1.795334	4	2	-0.018709	2

```
Missing Values
survived      0
pclass        0
sex           0
age          177
sibsp         0
parch         0
fare          0
embarked      2
class         0
who           0
adult_male    0
deck         688
embark_town   2
alive         0
alone         0
dtype: int64
```

```
Training Data Shape: (712, 7)
Testing Data Shape : (179, 7)
```

The Python environment was successfully set up and the dataset was pre-processed by handling missing values, encoding categorical data, performing feature scaling, and splitting the data into training and testing sets. The dataset is now ready for model training and analysis.